Phantasos-Lite — Experiment 3 Results Log: Llama-3.2-3B-Instruct

Reference doc for all numbers and findings from the K=16 Llama-3.2-3B-Instruct run (Tier B cross-family replication), completing the three-model set alongside Experiment 1 (Qwen2.5-7B-Instruct) and Experiment 2 (Qwen2.5-1.5B-Instruct). Every number below is taken directly from notebook output, not recalculated or rounded differently. All layer-dependent numbers use `PRIMARY_LAYERS=[7,13,21]`*, hardcoded to match both Qwen runs exactly after confirming* `num_hidden_layers==28` *for this model too.*

Model & setup

Model	`meta-llama/Llama-3.2-3B-Instruct`
hidden_size	3072
num_hidden_layers	28
`PRIMARY_LAYERS`	[7, 13, 21] — hardcoded to match Experiments 1 and 2
Personas (K)	16
I-set prompts	8
Checkpoint folder	`/content/drive/MyDrive/phantasos_K16lite_llama3B_checkpoints`
Access	Gated HF repo; required requesting/accepting Meta's license before `from_pretrained` would succeed

Persona set

Identical to Experiments 1 and 2: `assistant, physician, professor, teacher, interrogator, critic, pirate, pirate_story` (original 8) + `customer_service, drill_sergeant, poet, poet_story, therapist, comedian, librarian, diplomat` (new 8).

Cluster counts: anchor=1, expert=3 (physician, professor, librarian), warm=3 (customer_service, teacher, therapist), cold=4 (drill_sergeant, interrogator, diplomat, critic), playful=5 (pirate, poet, pirate_story, poet_story, comedian) — same cluster assignments as both prior runs.

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1. Environment checks — all passed

Check	Result
Hook output type	`torch.Tensor` (`HOOK_OUTPUT_IS_TUPLE = False`)
Digit token ids	`[16,17,18,19,20,21,22,23,24]` — all 9 distinct, single-token (re-verified fresh for this tokenizer, not copied from either Qwen run)
Chat template tail	correct — ends at `` `< ``
alpha=0 no-op (Check D1)	**True**
Per-position delta norm (Check D2)	last position ≈278.0 (expected ≈277.77), all earlier positions exactly 0.00

Known template difference from both Qwen runs: Llama's default chat template injects a fixed `Cutting Knowledge Date: December 2023 / Today Date: <run date>` preamble into the system block before the persona's own system prompt, which Qwen's template never does. This text is identical across every persona and prompt in this run, so it is not expected to bias the relative RDM structure the way a persona-specific confound would — but it is a genuine template-level difference from the other two models' conditioning, left unmodified (patching the template would be a riskier deviation than leaving native default behavior in place) and noted here for Methods.

2. S2 — persona expression (judge-scored, 0–3 scale)

persona	mean	n
assistant	0.38	16
physician	1.88	16
professor	1.88	16
teacher	2.00	16
interrogator	1.88	16
critic	2.06	16
pirate	2.75	16
pirate_story	2.00	16
customer_service	1.75	16
drill_sergeant	2.31	16
poet	2.31	16
poet_story	1.56	16
therapist	2.00	16
comedian	1.88	16
librarian	1.81	16
diplomat	1.38	16

- **S2 pass rate, excluding interrogator:** 0.85 (n=224) — threshold was ≥0.85, **meets it exactly**
- **S2 pass rate, all role personas:** 0.85 — meets threshold
- **Length–expression correlation:** r = −0.11 (negligible/slightly negative — length is not driving judged expression here, unlike Exp 1's modest positive confound of 0.43)
- **Comparison:** best S2 pass rate of all three experiments (7B: 0.69, 1.5B: 0.59, Llama-3B: **0.85**), and `assistant` baseline is the lowest of all three runs (0.38, vs 7B's 1.44 and 1.5B's 0.62) — the cleanest default-vs-persona separation observed in this project.

3. S5 — activation separability (leave-one-variant-out, layer 13)

- **Accuracy: 0.914** (chance = 0.0625 for K=16) — highest of all three experiments (7B: 0.906, 1.5B: 0.742, Llama-3B: **0.914**), despite this model being smaller than the 7B baseline.
- **Confusions (at layer 13):**
  - `assistant` → confused with `physician` (1/8 misclassified)
  - `pirate_story` → confused with `pirate` (5/8 misclassified)
  - `poet_story` → confused with `poet` (4/8 misclassified)
- **Interpretation:** confusions land precisely on the designed speaker-mode "surprise" pairs (the story-narrator variants vs. their first-person counterparts), the same clean pattern as Experiment 1 — not the generic-sink pattern Experiment 2 showed (where `therapist` absorbed misclassifications from four unrelated personas). Llama-3.2-3B's representation has fine-grained sub-cluster structure on par with, or better than, the much larger 7B model.

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4. Baseline RSA — activation vs. prompt text (embedding baseline)

Not the finding — this is the confound being controlled for.

layer	RSA(act, text)	perm-p
7	0.557	0.0002
13	0.419	0.0002
21	0.427	0.0002

Note the non-monotonic shape (dips at 13, ticks back up at 21) — different from both Qwen runs' strictly decreasing baseline RSA across depth.

- **Shuffled-label control RSA (layer 13):** −0.063 (expected ≈0 — passed)
- **Split-half noise ceiling (variant 0 vs. variant 1):**
  - layer 7: 0.913
  - layer 13: 0.948
  - layer 21: 0.968
- **Comparison:** dramatically higher ceiling than either Qwen run at every layer, especially at the shallow layer (0.913 vs. 7B's 0.769 and 1.5B's 0.529 at layer 7). Llama-3.2-3B's

activation geometry is far more internally consistent across prompt variants independent of the self-report question entirely.

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5. RDM-S validity (judge-free pairwise self-rating)

- 256/256 ratings collected (16×16, including diagonal self-ratings)
- **Low-confidence ratings (digit_mass < 0.8): 170 / 256 (66.4%)** — expected 0 or close to it; this is the one real finding of concern in this run, investigated live rather than left unexplained (see below).
- **Self-rating EV (expected ≈9):** mean = 7.16, min = 6.15

Investigation of the low-confidence ratings

The rating question already ends with an explicit, forceful instruction ("*Answer with a single digit only.*"), the same style of directive wording presumably responsible for both Qwen runs achieving 0/256 low-confidence ratings. Probing the model's actual top-predicted tokens at the generation point (via a synthetic top-k logit check across several plausible prompt endings) confirmed:

- When the prompt's generation point is immediately preceded by a space (`": "`, `" out of 9: "`), the model confidently predicts bare digit tokens — the exact ids captured in `DIGIT_IDS` — with high logits and correctly concentrated mass.
- Llama's actual chat-template generation point ends in **two newlines** (`<|start_header_id|>assistant<|end_header_id|>\n\n`), not a space — Qwen's template ends in a single newline. Under this structure, the model's top prediction is still frequently a digit, but with meaningfully more competing mass on hedging continuations ("How", "I", "(") than when a direct space precedes the digit position.

Conclusion: not a bug. `DIGIT_IDS` correctly captures the tokens the model actually uses when it does commit to a digit (confirmed directly via the top-k probe); the question wording is already maximally direct; the code is structurally identical to both Qwen runs, which produced clean 0/256 results under the same logic. The 66% low-confidence rate reflects a genuine, model-specific difference in terse single-token instruction-following between Llama-3.2-3B and both Qwen2.5 models under their respective templates' generation-point structure — left unmodified and reported honestly rather than patched, since no defensible code-level fix was identified.

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6. PRIMARY RESULT — activation ↔ self-report correspondence

This is the paper's headline finding.

layer	self-desc RSA (raw)	self-desc (partial-out text)	self-rating RSA (raw)	self-rating (partial-out text)
7	0.660 (p=0.0002)	0.563	0.440 (p=0.0004)	0.334
13	**0.817** (p=0.0002)	**0.777**	**0.717** (p=0.0002)	**0.678**
21	0.785 (p=0.0002)	0.737	0.695 (p=0.0002)	0.652

- **RDM-F / RDM-S channel agreement:** 0.667 (between Exp 1's 0.723 and Exp 2's 0.456 — reasonable given the self-rating channel's confidence issue noted in §5, yet still much closer to Exp 1's strong agreement than Exp 2's)
- **Pattern — genuinely different from both Qwen runs:** both channels peak at **layer 13**, not the deepest layer. Both Qwen models show strictly monotonic increasing RSA through depth, peaking at layer 21. Llama-3.2-3B peaks at the middle layer and dips slightly by the final layer — a real, architecture-dependent difference in *where* the correspondence is strongest, not just *how strong* it is.

Ceiling-normalized comparison (RSA ÷ split-half reliability), all three models

layer	self-desc 7B	self-desc 1.5B	self-desc Llama-3B	self-rating 7B	self-rating 1.5B	self-rating Llama-3B
7	0.603	0.788	0.723	0.554	0.448	0.482
13	0.736	0.925	**0.862**	0.777	0.711	**0.756**
21	0.750	0.784	0.811	0.806	0.802	0.718

Even after correcting for its much higher noise ceiling, Llama-3.2-3B's self-description correspondence remains strong at every layer (0.723/0.862/0.811), landing between the two Qwen models at most layers rather than being an outlier. Self-rating's ceiling-normalized correspondence is weaker relative to self-description at layer 21 specifically (0.718, the lowest of the nine cells in this table) — consistent with the confidence issue in §5 degrading that one channel rather than the underlying representational correspondence.

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7. Bugs and notes encountered this run (for Methods/reproducibility notes, not results)

- **Gated repo access:** `meta-llama/Llama-3.2-3B-Instruct` required requesting and being granted access on HuggingFace before loading would succeed (`GatedRepoError` on first

attempt) — resolved by accepting Meta's license on the model page with the same account as the `HF_TOKEN` secret; approval was near-instant.

- **`HF_TOKEN` login bug:** initial loading cell called `login(token=os.environ["HF_TOKEN"])` without ever setting `os.environ["HF_TOKEN"]` (and without importing `os` in that cell) — fixed to `login(token=HF_TOKEN)`, using the variable already fetched via `userdata.get`.
- **Stale comparison strings copy-pasted from the Qwen-7B notebook:** several `EXPECTED:` print lines referenced `hidden_size=3584`, `Qwen2.5-7B-Instruct` by name, "chance level (8 personas): 0.125", and "128 for the pilot" — all leftover text from earlier notebooks/pilot stages, not updated for this model or this K=16 scale. In every case the actual computed value (hidden_size=3072, n_layers=28, 256 jobs, etc.) was correct; only the printed comparison text was wrong. Cosmetic only, fixed for clarity where caught.
- **`tok` vs. `tokenizer` naming mismatch:** the Check A–D environment-validation block (copied verbatim from the original pilot protocol) referenced `tok` throughout, but the model-loading cell defined the tokenizer as `tokenizer` — fixed with a one-line alias (`tok = tokenizer`) rather than renaming every reference.
- **Silent multi-token digit risk in Check B:** the digit-token check printed a "MULTI-TOKEN, PROBLEM" warning for any non-single-token digit but did not actually assert on it, silently continuing with a partial/wrong token id regardless — added an explicit assert. (Not triggered in practice — all nine digits for this tokenizer are confirmed single-token.)
- **Duplicate, conflicting `digit_ids` cells:** an earlier standalone cell computed digits 0–8 with a leading space; the canonical Check B cell (matching the actual protocol used in both Qwen runs) computes digits 1–9 with no leading space. The two definitions disagreed; resolved by deleting the incorrect standalone cell rather than relying on cell-execution order to silently overwrite it.
- **Mid-run disconnect:** the laptop running the Colab session died during the Groq judge pass (145/256 complete at the time). On reconnect, the S2 description generation (already fully checkpointed at 256/256) and the Groq judge pass both resumed correctly from their saved checkpoints rather than restarting — the judge pass picked up at "145 already done" with no lost work, confirming the checkpoint-resume logic held up under an actual unplanned interruption, not just the planned layer-mismatch reruns from Experiment 2.
- **170/256 low-confidence self-ratings** — investigated in depth; see §5. Not a bug; reported as a genuine, model-specific limitation of the self-rating channel for this model.
- **Chat-template date-preamble injection** — see §1. Not a bug; a structural template difference from Qwen, assessed low-risk for the RSA finding and documented as a limitation.

8. Files on Drive (checkpoint folder)

`s2_descriptions.jsonl`, `s2_judge.jsonl`, `act_index.jsonl`, `act_vectors.npz`, `rdm_s_ratings.jsonl`, `lite_rdm_heatmaps.png`, `lite_mds.png` — all confirmed complete (256/256/256/768/256 respectively).

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9. One-paragraph summary

Llama-3.2-3B-Instruct, K=16 personas, replicates the project's headline finding across a different model family entirely (Tier B cross-family replication), not just a different scale within the same family: a strong, statistically significant correspondence between internal activation geometry and self-report that survives partialing out prompt-text wording (self-description RSA up to 0.817, self-rating up to 0.717, both at layer 13). Unlike both Qwen runs, the correspondence peaks at the *middle* layer rather than the deepest one — a real architecture-dependent difference in where the effect is strongest. S2 persona-expression (0.85 pass rate) and S5 activation separability (0.914, with confusions landing precisely on the designed surprise pairs) are both the best of all three experiments, and the activation representation itself is far more split-half-reliable than either Qwen model at every layer. The one genuine weakness is the self-rating channel's confidence: 66% of ratings fell below the pre-registered digit-mass threshold, traced through direct top-token probing to a real difference in how reliably Llama-3.2-3B commits to a terse single-digit answer under its own chat template's generation-point structure, compared to Qwen's near-perfect compliance under its own template — not a pipeline bug, and not something a code fix could resolve without deviating from the model's native formatting. Self-description remains clean and strong across all three models; self-rating is the channel most sensitive to model-family-specific instruction-following calibration, a conclusion the cross-family comparison specifically enables that two same-family runs alone could not have shown.